

THE THYROID GLAND

1. Location and Structure

In most mammals, the **thyroid gland** is located **caudal to the trachea**, usually around the **1st or 2nd tracheal ring**.

It is one of the most important endocrine glands for **metabolic regulation**.

The thyroid is made up of many circular structures called **thyroid follicles**.

Thyroid follicle

Each follicle consists of follicular cells arranged in a circle around a central space called the **follicular lumen**.

The lumen contains a thick, homogeneous substance called **colloid**.

Colloid = major storage site of thyroid hormones

Follicular cell appearance

The shape of follicular cells changes depending on their activity:

- **Cuboidal cells** → relatively inactive/basal secretion
- **Elongated cells** → actively stimulated to release thyroid hormone

2. Parafollicular Cells (C Cells)

Another important thyroid cell is the:

Parafollicular cell / C cell

These cells are located **outside the thyroid follicles**.

They produce:

Calcitonin

Calcitonin is important in the regulation of **calcium metabolism**.

Remember:

Follicular cells → T3 and T4

C cells → calcitonin

THYROID HORMONE SYNTHESIS

3. Thyroid Hormones Are Made From Tyrosine and Iodine

The thyroid hormones are unusual because they are made from:

- **Tyrosine**
- **Iodine**

They are also unusual because a large amount of the finished hormone is stored **outside the cells**, within the colloid.

The main thyroid hormones are:

- **T4 = thyroxine / tetraiodothyronine**
- **T3 = triiodothyronine**

They contain **4 and 3 iodine atoms**, respectively.

4. Thyroglobulin

Tyrosine is part of a very large protein called:

Thyroglobulin (Tg)

Thyroglobulin is produced inside the **thyroid follicular cells** and then secreted into the **follicular lumen/colloid**.

Therefore:

Follicular cell → thyroglobulin → colloid

The tyrosine residues in thyroglobulin provide the basic structure from which thyroid hormones are formed.

5. Iodide Uptake

Iodine from the diet is converted to **iodide** in the gastrointestinal tract.

The iodide is then transported through the blood to the thyroid.

Thyroid follicular cells actively **trap iodide** from the blood.

Because of this active transport:

Intracellular iodide concentration can become about 25–200 times higher than the extracellular concentration.

This allows the thyroid to collect enough iodide to make thyroid hormones.

6. Addition of Iodine to Tyrosine

Iodide moves across the **apical membrane** of the follicular cell toward the colloid.

It is then attached to tyrosine residues within thyroglobulin.

A tyrosine ring can receive up to **2 iodine molecules**.

One iodine:

Monoiodotyrosine (MIT)

Two iodines:

Diiodotyrosine (DIT)

7. Formation of T3 and T4

The iodinated tyrosine molecules are then **coupled together**.

T4 formation

DIT + DIT → T4

T4 is also called:

Tetraiodothyronine / thyroxine

It contains **4 iodine atoms**.

T3 formation

MIT + DIT → T3

T3 is called:

Triiodothyronine

It contains **3 iodine atoms**.

Easy memory:

DIT + DIT = T4

MIT + DIT = T3

8. Thyroperoxidase

An essential enzyme in thyroid hormone synthesis is:

Thyroperoxidase (TPO)

TPO works together with **hydrogen peroxide (H₂O₂)**.

It is responsible for:

- Attaching iodine to tyrosine residues of thyroglobulin
- Helping form **T3 and T4**

Simple pathway:

Tyrosine + iodine → MIT/DIT → coupling → T3/T4

Unique feature:

Thyroid hormones are the **only hormones that contain a halogen**, specifically **iodine**.

THYROID HORMONE STORAGE

9. Thyroid Hormones Are Stored Outside the Cells

Most endocrine hormones are stored inside cells.

The thyroid is unusual because its hormones are stored **outside the follicular cells**, in the **colloid**.

The hormones remain attached to **thyroglobulin** while they are stored.

Why is this important?

It gives the thyroid a **large reserve of thyroid hormone**.

This allows animals to continue producing adequate thyroid hormone for some time even during periods of **iodine deficiency**.

Remember:

Most endocrine glands → store hormone inside cells

Thyroid → stores hormone outside cells in colloid

RELEASE OF THYROID HORMONES

10. How T3 and T4 Are Released

Before the thyroid can release T3 and T4, the thyroglobulin containing the hormones must be brought **back into the follicular cell**.

Sequence:

Colloid thyroglobulin

- taken into follicular cell
 - fuses with lysosomes
 - lysosomal enzymes break down thyroglobulin
 - T3 and T4 are released
 - T3 and T4 leave the cell
 - enter surrounding/interstitial fluid and then blood
-

11. Role of Lysosomes

The thyroglobulin-containing vesicles fuse with **lysosomes** inside the follicular cell.

The lysosomal enzymes cleave:

- **MIT**
- **DIT**
- **T3**
- **T4**

from thyroglobulin.

The released **T3 and T4** then pass through the basal cell membrane into the surrounding tissue and bloodstream.

Because thyroid hormones are lipid-soluble, they can **freely cross the cell membrane**.

12. Recycling of Iodine and Tyrosine

The thyroid does not waste the iodine-containing building blocks.

After thyroglobulin is broken down:

MIT and DIT → deiodination → iodide + tyrosine

An enzyme called:

Iodotyrosine dehalogenase

removes iodine from MIT and DIT.

The recovered:

- **Iodide**
- **Tyrosine**

are recycled and used to make more thyroid hormone.

Important:

MIT + DIT → iodine and tyrosine recycled

T3 AND T4 PRODUCTION

13. The Thyroid Mainly Produces T4

The thyroid gland itself produces:

- **Mostly T4**
- **Small amounts of T3**

However, most of the body's **T3 is produced outside the thyroid**.

This happens when tissues convert **T4 → T3**.

14. Conversion of T4 to T3

The major tissues involved in converting T4 to T3 include:

- **Liver**
- **Kidneys**
- **Muscle**

The liver and kidneys have high concentrations of deiodinating enzymes.

However, because muscle makes up such a large portion of the body, it produces a large amount of T3 overall.

The important enzyme that removes iodine from the **outer ring** of T4 is:

5'-monodeiodinase

Reaction:

T4 → T3

T3 is an important biologically active thyroid hormone.

15. Reverse T3

T4 can also be converted into:

Reverse T3 (rT3)

In this case, iodine is removed from the **inner ring** instead of the outer ring.

T4 → reverse T3

Reverse T3 has **very little biological activity** compared with T3.

Importantly:

The thyroid gland itself does not produce reverse T3.

It is produced by **extrathyroidal tissues** through deiodination of T4.

T3/T4 CONVERSION SUMMARY

T4 → T3

- Iodine removed from the **outer ring**
- Produces biologically active T3
- Mainly occurs outside the thyroid

T4 → reverse T3

- Iodine removed from the **inner ring**
 - Produces a form with little biological activity
 - Also occurs outside the thyroid
-

TRANSPORT OF THYROID HORMONES IN BLOOD

16. Thyroid Hormones Are Transported Bound to Plasma Proteins

T3 and T4 are **lipid-soluble**, so they do not dissolve well in the water-based plasma. Therefore, most thyroid hormones travel in the blood attached to **plasma binding proteins**.

There is considerable **species variation** in the proteins that transport thyroid hormones.

17. Main Thyroid Hormone-Binding Proteins

There are three important plasma proteins:

A. Thyroxine-binding globulin (TBG)

TBG is generally the most important specific carrier.

It:

- Has **high affinity for T4**
- Also carries **T3**
- Has relatively **low capacity** because its concentration in plasma is low

Important veterinary point:

TBG has been found in domestic animals **except cats**.

B. Albumin

Albumin:

- Has **low affinity** for T3 and T4
- Has **high capacity**

Why?

Because albumin is present in **very large amounts in plasma**.

Therefore, albumin can transport a large amount of thyroid hormone despite its weaker binding.

In the absence of TBG:

Albumin becomes the most important carrier.

This is particularly relevant in species such as the **cat**, which lacks TBG.

C. Thyroxine-binding prealbumin

A third transport protein is:

Thyroxine-binding prealbumin

It:

- Specifically binds **T4**
- Has **intermediate affinity and capacity**
- Falls between TBG and albumin in its binding characteristics

The term "**prealbumin**" refers to how the protein moves during **electrophoresis**. It does **not** mean that it is a precursor used to make albumin.

18. Free vs Bound Thyroid Hormone

Most T3 and T4 in the blood is **protein-bound**.

Only a very small amount is **free**.

The **free hormone** is the biologically available form that can leave the blood and interact with target cells.

Human values:

- Free T4 ≈ **0.03%**
- Free T3 ≈ **0.3%**

Dog values:

Dogs have a somewhat larger free fraction because their plasma proteins have a lower affinity for thyroid hormones.

- Free T4 = **slightly <1%**
- Free T3 = **slightly >1%**

Key point:

Bound hormone = mainly transport/storage reservoir

Free hormone = available to interact with target tissues

19. Equilibrium Between Free and Bound Hormone

The amount of free and bound thyroid hormone can change depending on physiological or drug-related conditions.

There is a continuous equilibrium:

Bound hormone $\rightleftharpoons$ Free hormone

When free hormone is used by tissues, more hormone can be released from the protein-bound pool.

The body rapidly adjusts this balance to maintain an appropriate concentration of free hormone.

20. Factors That Can Change Free Thyroid Hormone Levels

The free/bound balance can be affected by:

- **Illness**
- **Drugs**
- Changes in plasma-binding proteins
- Increased **estrogen**, such as during pregnancy

When binding changes, the body can compensate by:

- **Reducing hormone metabolism**, and/or
- Increasing thyroid hormone production through **TSH stimulation**

These responses help maintain an appropriate amount of **free thyroid hormone**.

HIGH-YIELD SUMMARY

Thyroid structure

Thyroid follicle $\rightarrow$ follicular cells + colloid

Colloid = major storage site of thyroid hormones

C cells $\rightarrow$ calcitonin

Hormone synthesis

Iodide uptake $\rightarrow$ tyrosine in thyroglobulin $\rightarrow$ MIT/DIT $\rightarrow$ coupling $\rightarrow$ T3/T4

Remember:

MIT = 1 iodine

DIT = 2 iodines

MIT + DIT = T3

DIT + DIT = T4

TPO + H₂O₂ $\rightarrow$ iodination + coupling

Storage

T3/T4 are stored outside the follicular cells, attached to thyroglobulin in colloid.

This provides a **large hormone reserve** and helps protect against short-term iodine deficiency.

Release

Thyroglobulin from colloid $\rightarrow$ follicular cell $\rightarrow$ lysosomes $\rightarrow$ T3/T4 released $\rightarrow$ blood

MIT/DIT $\rightarrow$ deiodination $\rightarrow$ iodine + tyrosine recycled

Peripheral conversion

Thyroid $\rightarrow$ mostly T4

Peripheral tissues $\rightarrow$ T4 $\rightarrow$ T3

T4 $\rightarrow$ T3 = outer-ring deiodination

T4 $\rightarrow$ reverse T3 = inner-ring deiodination

Reverse T3 = little biological activity

Blood transport

Because T3 and T4 are lipid-soluble:

Most = protein-bound

Very little = free and biologically available

Main transport proteins:

TBG $\rightarrow$ high affinity, low capacity

Albumin $\rightarrow$ low affinity, high capacity

Thyroxine-binding prealbumin $\rightarrow$ intermediate

Important species point:

Cats do not have TBG.

MOST IMPORTANT CONCEPTS TO MEMORIZE

1. Follicular cells make T3/T4; C cells make calcitonin.

2. Thyroid hormones are made from tyrosine + iodine.

3. TPO is essential for iodination and coupling.

4. T3 = MIT + DIT.

5. T4 = DIT + DIT.

6. The thyroid stores large amounts of T3/T4 in colloid attached to thyroglobulin.

7. The thyroid produces mostly T4; most T3 is produced in peripheral tissues from T4.

8. Reverse T3 has little biological activity.

9. Most T3/T4 in blood is protein-bound; the free fraction is the biologically available form.

10. TBG has high affinity/low capacity, whereas albumin has low affinity/high capacity.

THYROID HORMONE METABOLISM, ACTIONS, AND REGULATION

1. Main Ways Thyroid Hormones Are Metabolized

The main way thyroid hormones are broken down is by **removing iodine (deiodination)**.

Another, less common pathway is **conjugation with sulfates and glucuronides**, mainly in the **liver and kidneys**.

A third pathway involves changing the **alanine part** of the hormone through:

- **Transamination**
 - **Decarboxylation**
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2. Deiodination

Deiodination = removal of iodine from thyroid hormones.

The enzymes involved in producing T3 and reverse T3 also help break down thyroid hormones:

- **5'-deiodinase**
- **5-deiodinase**

These enzymes can remove iodine from either the **3 or 5 position** of the hormone's phenolic rings.

Important tissues for thyroid hormone breakdown:

- **Skeletal muscle**
- **Liver**
- **Kidneys**

Biological activity of deiodinated products

Most deiodinated thyroid hormone products have **little or no important metabolic activity**.

The major exception is:

T4 → T3

T3 is biologically active and is an important thyroid hormone.

3. Conjugation

Thyroid hormones can also be inactivated by attaching:

- **Sulfates**
- **Glucuronides**

These conjugates are produced mainly in:

- **Liver**
- **Kidneys**

However:

Conjugation is less common than deiodination as a pathway for thyroid hormone metabolism.

4. Other Metabolic Pathways

Another less important pathway changes the **alanine portion** of thyroid hormones.

This can happen through:

- **Transamination**
- **Decarboxylation**

These reactions contribute to further breakdown and inactivation of thyroid hormones.

5. Excretion of Thyroid Hormone Metabolites

Most metabolized thyroid hormones are eliminated through the **urine**.

Mainly in urine:

- **Deiodinated forms**
- **Conjugated forms**

However, some **unmetabolized thyroid hormones** are secreted into **bile** and eventually eliminated in the **feces**.

The conjugates in the intestine can then be broken down, releasing **iodide**.

The iodide can be absorbed again and returned to the body through the:

Enterohepatic cycle

Simple flow:

Thyroid hormone → liver metabolism → bile → intestine → iodide released → reabsorbed → returned to circulation

6. Iodide Recycling: Humans vs Dogs

Humans are more efficient than dogs at recovering iodide through:

- **Intrathyroidal recycling**
- **Enterohepatic recycling**

This means humans are generally better at conserving and reusing iodine.

HALF-LIFE OF THYROID HORMONES

7. Thyroid Hormones Have Long Half-Lives in Humans

Thyroid hormones remain in the body much longer than many other hormones.

In humans:

T3 → half-life ≈ 1 day

T4 → half-life ≈ 6–7 days

Most other hormones have half-lives of only **minutes to hours**.

8. Why T4 Lasts Longer Than T3

One important reason thyroid hormones have long half-lives is that much of them is **bound to plasma proteins**.

Protein binding protects the hormones from being rapidly broken down.

T4 has a longer half-life than T3 because:

T4 binds more tightly to plasma proteins → less free T4 → slower degradation → longer half-life

9. Important Veterinary Species Difference

The half-life of T4 is much shorter in some domestic animals.

In:

- **Dogs**
- **Cats**

T4 has a half-life of **less than 24 hours**.

This is much shorter than the human T4 half-life of **6–7 days**.

ACTIONS OF THYROID HORMONES

10. Thyroid Hormones Control Basal Metabolism

Thyroid hormones are considered the **primary regulators of basal metabolic activity**.

They are important for the normal metabolic activity of **almost all tissues**.

Because thyroid hormones affect many tissues simultaneously, their effects are difficult to separate into completely independent functions.

11. How Thyroid Hormones Act on Cells

Although thyroid hormones are made from **amino acids**, they behave much like **lipid-soluble hormones**.

They can:

Cross the cell membrane → enter the cell → interact with intracellular receptors

Their main action is believed to involve the **nucleus**, where they influence the transcription of **mRNA**.

Thyroid hormone receptors have also been identified on **mitochondria**, suggesting that thyroid hormones may have some direct effects there as well.

12. Calorigenic Effect

One of the most important actions of thyroid hormones is increasing:

- **Oxygen consumption**
- **Heat production**

This is called the:

Calorigenic effect

Simple relationship:

↑ Thyroid hormone → ↑ oxygen consumption → ↑ heat production

The **mitochondria** are one important site involved in this effect.

CARBOHYDRATE METABOLISM

13. Effects on Carbohydrates

Thyroid hormones affect carbohydrate metabolism in several ways.

They:

- **Increase intestinal absorption of glucose**
- Help glucose enter **muscle and fat cells**
- Make cells more responsive to **insulin-mediated glucose uptake**

Effect on glycogen

Small amounts of thyroid hormone can:

↑ Glycogen formation

Higher amounts can promote:

↑ Glycogenolysis

where glycogen is broken down to release glucose.

PROTEIN METABOLISM AND GROWTH

14. Role in Growth and Development

Thyroid hormones work together with **growth hormone (GH)** and are essential for:

- **Normal growth**
- **Normal development**

Part of their effect is due to increasing:

- **Amino acid uptake by tissues**
- Activity of enzymes involved in **protein synthesis**

Thus:

Thyroid hormone + GH → normal growth and development

LIPID METABOLISM

15. Effects on Lipids

Thyroid hormones affect many aspects of fat metabolism, with an important effect on:

Lipolysis = breakdown of fat

They also tend to **decrease plasma cholesterol**.

This appears to happen because thyroid hormones:

1. Increase cellular uptake of **LDLs**
2. Increase breakdown of **LDL and cholesterol**

Important clinical connection:

Thyroid hormone deficiency → decreased cholesterol clearance → hypercholesterolemia

Therefore:

Hypercholesterolemia is a characteristic finding of thyroid hormone deficiency.

16. Catabolic Effects

The effects of thyroid hormones on:

- Carbohydrates
- Proteins
- Lipids

are often described as **catabolic**, especially when thyroid hormone activity is excessive.

"Catabolic" means promoting the **breakdown or use of stored nutrients**.

NERVOUS SYSTEM EFFECTS

17. Thyroid Hormones and the Nervous System

Thyroid hormones are important for the **normal development of the nervous system**.

This is especially important during:

- **Fetal development**

- **Neonatal development**

Insufficient thyroid hormone during development can impair **mental and nervous system development**.

In adults, inadequate thyroid hormone can be associated with:

- **Mental dullness**
- **Lethargy**
- Reduced mental activity

Therefore, adequate thyroid hormone is important for **normal CNS function** throughout life.

18. Thyroid Hormones and the Sympathetic Nervous System

Thyroid hormones increase the effects of the **sympathetic nervous system**.

They do this partly by increasing the number or activity of:

β-adrenergic receptors

These receptors respond to catecholamines such as:

- **Epinephrine**
- **Norepinephrine**

Therefore:

↑ Thyroid hormone → ↑ β-adrenergic responsiveness → stronger sympathetic effects

CARDIOVASCULAR EFFECTS

19. Effects on the Heart

Thyroid hormones increase:

- **Heart rate**
- **Force of cardiac contraction**

This occurs partly because they increase the heart's responsiveness to **catecholamines**.

They promote the expression of **β-adrenergic receptors**, making cardiac tissue more responsive to sympathetic stimulation.

20. Effects on Blood Pressure

Thyroid hormones can increase:

Systolic blood pressure

while generally causing little change in:

Diastolic blood pressure

This results in:

↑ Cardiac output

Thyroid hormone activity is therefore important for maintaining normal:

- Cardiac contraction
 - Cardiac function
 - Transmission of nerve impulses in cardiac tissue
-

These effects are especially obvious when thyroid hormone levels are excessively high.

THYROID HORMONES AND DEVELOPMENT

21. Amphibian Metamorphosis

A classic demonstration of thyroid hormone action comes from amphibians.

Normal:

Thyroxine → tadpole develops into frog

Without a thyroid:

Thyroidectomy → tadpole remains large and does not normally undergo metamorphosis

This dramatic effect is particularly characteristic of **amphibians**.

In other animals, thyroid hormones do not cause such obvious metamorphosis, but they are still important for many **subtle developmental and differentiation processes**.

22. Overall Role of Thyroid Hormones

Although thyroid hormone effects are often described based on what happens when hormone levels are:

- Too low
- Too high

it is better to remember that thyroid hormones are fundamentally needed for the **normal metabolic activity of almost every tissue in the body**.

REGULATION BY TSH

23. TSH Is the Main Regulator of the Thyroid Gland

Thyroid-stimulating hormone (TSH), also called **thyrotropin**, is the major hormone controlling thyroid gland activity.

TSH stimulates the thyroid through a signaling pathway involving:

TSH receptor → cAMP → protein kinase activation → thyroid gland stimulation

This promotes thyroid hormone synthesis and release.

24. Negative Feedback of Thyroid Hormones

Thyroid hormones regulate their own production through **negative feedback**.

When T3 and T4 levels increase:

↑ T3/T4 → ↓ TRH from hypothalamus → ↓ TSH from anterior pituitary → ↓ thyroid activity

When thyroid hormone levels decrease:

↓ T3/T4 → less negative feedback → ↑ TRH → ↑ TSH → ↑ thyroid activity

Therefore, the system continuously adjusts thyroid hormone production to maintain appropriate levels.

OVERALL THYROID AXIS

When thyroid hormone is LOW:

↓ T3/T4

↓ negative feedback

→ ↑ **TRH** from hypothalamus

→ ↑ **TSH** from anterior pituitary

→ thyroid gland stimulated

→ ↑ **T4 and T3**

When thyroid hormone is HIGH:

↑ T3/T4

→ stronger negative feedback

→ ↓ **TRH**

→ ↓ **TSH**

→ reduced thyroid stimulation

→ ↓ **T3/T4 production**

HIGH-YIELD SUMMARY

Thyroid Hormone Metabolism

Main pathway = deiodination

Important tissues:

Liver + kidney + skeletal muscle

Other pathways:

- Sulfate conjugation
- Glucuronide conjugation
- Transamination
- Decarboxylation

Excretion:

Deiodinated/conjugated forms → urine

Unmetabolized hormone → bile → feces

Some iodide is recovered through the **enterohepatic cycle**.

Half-Life

Humans:

T3 ≈ 1 day

T4 ≈ 6–7 days

Dogs and cats:

T4 <24 hours

Protein binding helps prolong thyroid hormone half-life.

Major Actions of Thyroid Hormones

Metabolism

↑ **Basal metabolic rate**

↑ **Oxygen consumption**

↑ **Heat production**

Carbohydrates

- ↑ Glucose absorption
- ↑ Glucose uptake
- ↑ Insulin-mediated glucose uptake
- Small amounts → ↑ glycogen formation
- Large amounts → ↑ glycogenolysis

Proteins and growth

- Work with **GH**
- Essential for normal growth and development
- ↑ Amino acid uptake
- Support protein synthesis

Lipids

- ↑ Lipolysis
- ↓ Plasma cholesterol
- ↑ LDL uptake and degradation

Thyroid deficiency → hypercholesterolemia

Nervous system

- Essential for fetal/neonatal CNS development
- Adequate thyroid hormone supports normal adult CNS function

Cardiovascular system

- ↑ Heart rate
- ↑ Contractility
- ↑ β-adrenergic responsiveness
- ↑ Systolic pressure
- ↑ Cardiac output

Development

Thyroxine → amphibian metamorphosis

THYROID REGULATION

Hypothalamus → TRH

↓

Anterior pituitary → TSH

↓

Thyroid → T4 + T3

↓

T3/T4 –| TRH + TSH

This is **negative feedback**.

Most important concepts to memorize

1. **Deiodination is the main pathway of thyroid hormone metabolism.**
2. **T4 → T3 produces an active hormone, while most other deiodinated products have little activity.**
3. **Thyroid hormones have long half-lives in humans because of strong plasma protein binding.**
4. **Thyroid hormones increase basal metabolic activity, oxygen consumption, and heat production.**
5. **Thyroid hormones increase tissue responsiveness to catecholamines by increasing β-adrenergic receptor activity.**
6. **Thyroid hormone deficiency commonly causes hypercholesterolemia.**
7. **TSH is the main regulator of thyroid gland activity.**
8. **T3 and T4 provide negative feedback to both the hypothalamus and anterior pituitary.**

THYROID DISORDERS

1. Goitrogens and Goiter

What is a goiter?

A **goiter** is an enlargement of the thyroid gland that usually develops when the thyroid cannot produce enough thyroid hormone.

One major cause is **iodine deficiency**.

Iodine deficiency

If the diet does not provide enough iodine:

↓ **Iodine** → ↓ **thyroid hormone synthesis** → **thyroid stimulation increases** → **thyroid enlarges** → **goiter**

In many areas, iodine deficiency has been greatly reduced by using **iodized salt**.

2. Goitrogenic Plants and Compounds

Some plants contain substances called **goitrogens**, which interfere with thyroid hormone production.

Important examples include **cruciferous plants**:

- Cabbage
- Kale
- Rutabaga
- Turnip
- Rapeseed

These plants contain **progoitrin**, which is converted to **goitrin** in the digestive tract.

Goitrin

Goitrin interferes with the **organic binding of iodine** during thyroid hormone synthesis.

Other goitrogenic feeds contain **thiocyanates**.

Thiocyanates

Thiocyanates interfere with the thyroid's ability to **trap/actively take up iodide**.

Therefore:

Thiocyanates → ↓ **iodide uptake**

Goitrin → ↓ **organic binding of iodine**

Both can reduce thyroid hormone production and potentially cause **goiter and hypothyroidism**.

3. Effect of Excess Iodine

Increasing dietary iodine can sometimes help overcome the effects of **thiocyanates**.

However, excess iodine is **much less effective against goitrin**.

So:

Extra iodine → **may partially overcome thiocyanate effects**

Extra iodine → **has less effect against goitrin**

4. Antithyroid Drugs

Understanding how goitrogens interfere with thyroid hormone synthesis helped lead to the development of drugs used to treat **hyperthyroidism**.

The most potent group mentioned is the:

Thiocarbamides

- **Thiourea**
- **Thiouracil**

Other antithyroid compounds include:

- **Sulfonamides**
- **p-Aminosalicylic acid**
- **Phenylbutazone**
- **Chlorpromazine**

These compounds reduce thyroid hormone production through interference with thyroid hormone synthesis.

HYPOTHYROIDISM IN DOGS

5. Canine Hypothyroidism

Hypothyroidism is one of the most important endocrine diseases in **dogs**.

The most common cause of **primary hypothyroidism** is:

Lymphocytic thyroiditis

6. Causes of Hypothyroidism

A. Primary hypothyroidism

The problem is directly within the **thyroid gland**.

Common or possible causes include:

- **Lymphocytic thyroiditis**
- **Thyroid dysgenesis** = abnormal development of the thyroid
- **Dyshormonogenesis** = abnormal thyroid hormone production
- **T4 transport defects**

- **Goitrogens**
- Rarely, **iodine deficiency**

B. Secondary hypothyroidism

The problem is at the level of the **pituitary gland**.

Possible causes include:

- **Pituitary tumors**
- **Radiation therapy**
- **Glucocorticoid exposure**

Glucocorticoids may be either:

- Produced by the body (**endogenous**), or
- Given as medication (**exogenous**)

C. Tertiary hypothyroidism

The problem occurs at the level of the **hypothalamus**.

It may be:

- **Acquired**, such as from hypothalamic tumors
- **Congenital**, due to defects involving **TRH** or the **TRH receptor**

Easy way to remember:

Primary = thyroid problem

Secondary = pituitary problem

Tertiary = hypothalamus problem

7. Dogs at Higher Risk

Hypothyroid dogs show a breed and age predisposition.

Age

High-risk breeds may develop signs as early as:

2–3 years

Lower-risk breeds often develop signs around:

4–6 years

Breeds reported to have increased risk include:

- English Setters
- Eurasiers
- Spanish Water Dogs
- Irish Red and White Setters
- Dalmatians
- Boxers
- Shetland Sheepdogs
- Tibetan Terriers
- Kuvasz
- German Wirehaired Pointers

- Rhodesian Ridgebacks
- Other susceptible breeds

CLINICAL SIGNS OF CANINE HYPOTHYROIDISM

8. General Clinical Signs

The onset is usually:

Slow and subtle

The most common signs are:

- **Lethargy**
- **Obesity**

9. Skin and Hair Changes

Skin problems are among the most common findings after lethargy and obesity.

Classic finding:

Symmetrical truncal or tail-head alopecia

Other common changes include:

- **Thickened skin**
- **Dull, dry hair**
- **Poor hair regrowth after clipping**
- **Retained puppy hair coat**

The skin may become thick because of **myxedematous accumulation in the dermis**.

Myxedema

This refers to the accumulation of hydrophilic substances/fluid in tissues associated with severe thyroid hormone deficiency.

10. Cardiovascular Signs

Possible cardiovascular changes include:

- **Bradycardia**
- **Decreased cardiac contractility**
- **Atherosclerosis**

However, these are **uncommon as the main presenting complaint**.

11. Neuromuscular and Nervous System Signs

These are also relatively uncommon.

Possible signs include:

- **Myopathy**
- **Megaesophagus**
- **Neuropathies**
- **Facial nerve paralysis**
 - Bilateral or unilateral

- Vestibular disease
- Lower motor neuron disorders

12. Myxedema Coma

Myxedema coma is rare in dogs.

It is associated with:

- Severe hypothyroidism
- **Myxedematous fluid accumulation in the brain**
- Severe **hyponatremia**

It is a serious manifestation of thyroid hormone deficiency.

13. Reproductive Signs

Female dogs may develop reproductive problems such as:

- **Prolonged interestrous interval**
- **Silent heat**
- Delivery of **weak puppies**
- Delivery of **stillborn puppies**

14. Other Clinical Signs

Less common findings include:

- **Corneal lipid deposits**
- **Constipation**
- Other gastrointestinal disturbances

CLINICOPATHOLOGICAL FINDINGS

15. Anemia

Approximately **25–30% of hypothyroid dogs** may have anemia.

Possible causes include:

- ↓ **Erythropoietin**
- ↓ **Bone marrow activity**
- ↓ **Serum iron**
- ↓ **Iron-binding capacity**

The anemia is usually mild and may be **normocytic and normochromic**.

16. Hypercholesterolemia

Approximately **75% of hypothyroid dogs** develop:

Hypercholesterolemia

This occurs because of altered lipid metabolism, including:

- ↓ Fecal cholesterol excretion
- ↓ Conversion of lipids to bile acids

Important connection:

Hypothyroidism → ↓ lipid clearance → ↑ blood cholesterol

17. Hyponatremia

Mild **hyponatremia** has been reported in about **30% of hypothyroid dogs** in one study.

Possible mechanisms include:

- Increased total body water
- Reduced renal excretion of water
- Retention of water associated with hydrophilic deposits in tissues

18. Increased Creatine Phosphokinase

An unusual laboratory finding is:

↑ **Serum creatine phosphokinase (CK/CPK)**

This may be related to **hypothyroid myopathy**.

DIAGNOSIS OF CANINE HYPOTHYROIDISM

19. Main Diagnostic Tests

Diagnosis should not depend on one test alone.

It should combine:

- **Signalment**
- **History**
- **Physical examination**
- **Clinicopathological findings**
- **Thyroid function tests**

Important thyroid tests include:

- **Total T4 (TT4)**
- **Total T3 (TT3)**
- **Free T4 (FT4)**
- **Endogenous canine TSH**

20. Why T4 Results Can Be Misleading

Many factors can affect serum T4.

These include:

- **Age**
- **Breed**
- **Environmental temperature**
- **Body temperature**
- **Diurnal rhythm**
- **Obesity**
- **Malnutrition**
- **Concurrent illness**

- Drugs

Breed example

Sighthounds may naturally have lower TT4 and FT4 than other dogs.

Obesity

Obese dogs may have **mildly increased TT4**.

Puppies

Puppies may have TT4 concentrations approximately:

2–5 times higher than adult dogs

There is also an age-related decrease in:

- Serum TT4
- Response to TSH stimulation

21. Euthyroid Sick Syndrome

Euthyroid sick syndrome occurs when a dog has another illness that causes thyroid hormone changes even though the thyroid gland itself is not truly failing.

Typical changes include:

↓ **TT4**

and often:

↑ **reverse T3**

Diseases that can produce this pattern include:

- **Diabetes mellitus**
- **Chronic renal failure**
- **Hepatic insufficiency**
- **Infections**

Important:

A low TT4 **does not automatically mean hypothyroidism**.

Other illnesses must be considered.

22. Drugs That Can Lower TT4

Several drugs can decrease basal TT4 concentrations, including:

- **Anesthetics**
- **Phenobarbital**
- **Primidone**
- **Diazepam**
- **Trimethoprim-sulfa**
- **Quinidine**
- **Phenylbutazone**
- **Salicylates**
- **Glucocorticoids**

Therefore, medication history is important when interpreting thyroid tests.

23. Free T4

Free T4 (FT4) means the T4 that is **not bound to plasma proteins**.

In human medicine, FT4 is useful for distinguishing:

True hypothyroidism vs euthyroid sick syndrome

The preferred method is:

Equilibrium dialysis = gold standard

Other methods include:

Analog immunoassays

In humans, one FT4 measurement has approximately **90% diagnostic accuracy**.

In dogs, early studies suggested:

- **FT4 by equilibrium dialysis ≈ 90% accurate**
- Other FT4 assays were no better than TT4

However, **glucocorticoids can decrease both FT4 and TT4 in dogs**.

24. Endogenous TSH

Endogenous TSH helps evaluate the **feedback relationship between the thyroid, pituitary, and hypothalamus**.

In primary thyroid failure:

↓ **T4/FT4** → **pituitary detects low thyroid hormone** → ↑ **TSH**

Therefore:

Low FT4 + high TSH → **strongly suggests primary hypothyroidism**

The increase in TSH becomes especially pronounced as FT4 decreases, making TSH a sensitive indicator of early thyroid failure.

Important veterinary limitation

TSH should **not be used alone** to evaluate thyroid function in animals.

Approximately:

20–30% of hypothyroid dogs can have normal TSH concentrations

even though T3 and T4 are decreased.

25. Thyroglobulin Autoantibody (TgAA)

The **antithyroglobulin autoantibody test (TgAA)** can detect evidence of autoimmune thyroid disease.

It is useful for:

- Identifying thyroid pathology **before obvious hypothyroidism develops**
- Detecting dogs with hereditary thyroid disease
- Screening dogs before **breeding**

26. TSH Stimulation Test

For many years, the **TSH stimulation test** was considered the gold standard for diagnosing canine hypothyroidism.

However, it has important limitations.

It does **not reliably distinguish**:

- Early hypothyroidism
- Euthyroid sick syndrome

It also does not identify:

- **Secondary hypothyroidism**
- **Tertiary hypothyroidism**

Another major problem is that **exogenous bovine TSH is no longer commercially available**.

27. Other Thyroid Tests

Other tests include:

- **TRH stimulation test**
- **Thyroid scan/scintigraphy**
- **Thyroid biopsy**

However, each has disadvantages such as:

- Expense
- Inaccuracy
- Invasiveness

28. Practical Diagnostic Approach

A practical approach described in the notes is:

First:

Measure:

TT4 + TSH

Then:

Measure:

FT4 by equilibrium dialysis

Interpretation

All 3 abnormal:

TT4 abnormal + TSH abnormal + FT4 abnormal

→ Hypothyroidism is likely.

Two of the three abnormal:

Possible:

Primary hypothyroidism

- ↑ TSH
- ↓ FT4

OR

Secondary hypothyroidism

- ↓ TSH
- ↓ FT4

Only one abnormal:

Do not immediately diagnose hypothyroidism.

Reevaluate the dog in 3–6 months.

FELINE HYPERTHYROIDISM

29. General Information

Hyperthyroidism is the **most common endocrinopathy in cats**.

The most common cause is:

Adenomatous hyperplasia of the thyroid gland

It usually affects:

- **Middle-aged to older cats**

There is:

- No major breed predisposition
- No major sex predisposition

Although the incidence is reported to be higher in **purebred cats than mixed-breed cats**.

30. Dietary Iodine and Feline Hyperthyroidism

Commercial/balanced pet foods normally provide enough iodine.

However, the iodine content of foods can vary considerably.

Changes in dietary iodine have been studied as possible risk factors.

Important distinction:

Acute changes in dietary iodine can alter **FT4**.

However, with chronic changes, the thyroid usually **adapts**, so chronic dietary iodine variation is unlikely by itself to explain feline hyperthyroidism.

31. Goitrogens and Feline Hyperthyroidism

Goitrogens can clearly cause **hypothyroidism**.

There has also been a theory that chronic exposure to goitrogens could eventually contribute to **toxic nodular goiter and hyperthyroidism**.

This remains a proposed explanation rather than a clearly established cause.

32. Soy Isoflavones

Soy proteins contain **isoflavones**, including:

- **Genistein**
- **Daidzein**

These compounds have been found in many dry cat foods.

They have been suggested as possible thyroid-disrupting compounds because some studies show that they can interfere with:

- **Thyroperoxidase**
- **5'-deiodinase**

Study in cats

In a study of 18 healthy cats eating a soy diet containing **400 mg isoflavones/kg diet**:

- TT4 increased modestly but significantly

- FT4 increased modestly but significantly
- T3 did not change

However, human research has generally not shown harmful effects of soy isoflavones on thyroid function when soy is part of a **balanced diet with adequate iodine**.

Important interpretation:

The evidence linking soy isoflavones to feline hyperthyroidism is **inconsistent**.

33. Canned Cat Food and BADGE

Canned cat foods have been associated with feline hyperthyroidism in several epidemiological studies.

One suspected goitrogen is:

Bisphenol-A-diglycidyl ether (BADGE)

BADGE was used in the lining of easy-open **pop-top cans**.

The proposed mechanism is:

BADGE from can lining → possible leaching into food → ingestion by cat

Cats may be particularly susceptible because they have a **reduced ability to detoxify substances through hepatic glucuronidation**.

Bisphenol A has also been shown in experimental studies to:

- Reduce T3 binding
- Increase TSH
- Cause hyperthyroidism/goiter in some experimental animals and some humans

However, rodent studies suggest a **large safety margin**.

Important:

An epidemiological association does **not prove cause and effect**.

Although more than **90% of cats in the United States** eat commercial pet food as their main food source, only a relatively small proportion develop hyperthyroidism.

34. Pathology of Feline Hyperthyroidism

Feline hyperthyroidism resembles **toxic nodular goiter** in humans.

It is characterized by:

Autonomous growth of thyroid follicles

This means the thyroid tissue can grow and produce hormones **without normal dependence on TSH**.

35. TSH Receptor and G-Protein Signaling

Normally:

TSH → TSH receptor → G protein → cAMP → thyroid hormone production + cell growth

In toxic nodular goiter, abnormalities can occur in this signaling system.

Possible changes include:

- **Increased stimulatory G-proteins**

- **Decreased inhibitory G-proteins**

- Mutations in the **TSH receptor** that keep it activated even without TSH

Similar abnormalities have been investigated in hyperthyroid cats.

In some cats, **activating mutations of the TSH receptor** may contribute to disease.

Hyperthyroid cats may also have decreased expression of **inhibitory G-proteins**.

Important difference from human Graves disease:

Feline hyperthyroidism does **not appear to be caused by stimulating antibodies against the TSH receptor**.

36. Environmental Risk Factors

Several environmental factors have been investigated.

Cat litter

Use of cat litter has been associated with increased hyperthyroidism risk.

However, this may simply reflect **indoor living**.

Indoor cats tend to:

- Live longer
- Therefore have more opportunity to develop hyperthyroidism

Pesticides and herbicides

These have been associated with thyroid abnormalities in other species.

Flea-control products

Some studies found an association between flea-control products and feline hyperthyroidism.

However:

No specific product or ingredient was consistently identified as the cause.

37. Brominated Flame Retardants

Brominated flame retardants (BFRs) have also been proposed as possible environmental goitrogens/carcinogens.

One group of interest is:

Polybrominated diphenyl ethers (PBDEs)

These compounds were detected in cats at serum levels reported to be **10–400 times higher than levels found in humans**.

Possible reasons include:

- Indoor living
- Frequent grooming
- Exposure to furniture and carpets
- Smaller body size

However, this remains a proposed association, not proof of causation.

CLINICAL SIGNS OF FELINE HYPERTHYROIDISM

38. Hypermetabolic Signs

Hyperthyroidism produces a state of **hypermetabolism**.

Common clinical signs include:

- **Polyphagia**
- **Weight loss**
- **Vomiting**
- **Diarrhea**
- **Polydipsia**
- **Polyuria**

A common pattern is:

Increased appetite + weight loss

39. Sympathetic Nervous System Signs

Hyperthyroidism also increases sympathetic activity.

Typical signs include:

- **Hyperactivity**
 - **Tachycardia**
 - **Pupillary dilation**
 - **Behavioral changes**
-

40. Long-Term Complications

Long-standing hyperthyroidism can lead to:

Cardiovascular

- **Hypertrophic cardiomyopathy**
- **High-output heart failure**

General

- **Cachexia** = severe loss of body condition/muscle

Severe chronic disease may eventually be fatal.

CLINICOPATHOLOGICAL FINDINGS

41. Erythrocytosis and Stress Leukogram

Hyperthyroid cats may develop:

Erythrocytosis

and a:

Stress leukogram

The stress leukogram may include:

- **Neutrophilia**
- **Lymphocytosis**

These changes are related to increased circulating **catecholamines**.

42. BUN and Creatinine

Increased breakdown of muscle may cause:

↑ **BUN**

without a similar increase in creatinine.

Why?

Hyperthyroidism also increases:

Glomerular filtration rate (GFR)

The increased GFR may temporarily **mask underlying kidney disease**.

Therefore, a seemingly normal creatinine does not necessarily mean that renal function is normal.

43. Urinalysis

The effect on urinalysis is variable.

However, many hyperthyroid cats have:

↓ **Urine specific gravity**

especially when they have **polyuria**.

44. Liver Enzymes

The increased metabolic rate causes **liver hypermetabolism**.

As a result:

ALT and AST may be increased.

Increased liver enzyme activities occur in approximately:

80–90% of hyperthyroid cats

45. Cholesterol

Hyperthyroidism can cause:

↓ **Serum cholesterol**

This is mainly due to:

↑ **Hepatic clearance of cholesterol**

rather than reduced cholesterol synthesis.

DIAGNOSIS OF FELINE HYPERTHYROIDISM

46. Total T4

The usual initial diagnostic test is:

Total T4 (TT4)

An increased TT4 strongly supports hyperthyroidism in a compatible cat.

47. Free T4

As cats are diagnosed at earlier stages, **FT4** can be particularly useful for detecting:

Early or occult hyperthyroidism

However, FT4 should generally be interpreted together with clinical findings and other thyroid tests.

48. T3 Suppression Test

A **T3 suppression test** can help confirm hyperthyroidism in borderline cases.

In a normal cat:

Administering T3 causes negative feedback:

Exogenous T3 → ↓ TSH → ↓ thyroid activity → ≥50% decrease in T4

So:

Normal cat → T4 decreases by at least 50%

In a hyperthyroid cat:

The thyroid gland functions more autonomously and does not respond normally to the negative-feedback effect.

Therefore:

Hyperthyroid cat → T4 does not decrease appropriately

This makes the test useful for **borderline cases**.

49. Thyroid Scintigraphy

Thyroid scintigraphy can be used to visualize thyroid tissue that is **hyperfunctional**.

It helps identify areas of increased thyroid activity.

HIGH-YIELD COMPARISON

Canine Hypothyroidism vs Feline Hyperthyroidism

Feature	Canine Hypothyroidism	Feline Hyperthyroidism
Hormone state	↓ Thyroid hormone	↑ Thyroid hormone
Common species	Dogs	Cats
Typical age	Adult dogs	Middle-aged/older cats
Common cause	Lymphocytic thyroiditis	Adenomatous hyperplasia
Metabolic state	Hypometabolic	Hypermetabolic
Weight	Weight gain/obesity	Weight loss despite ↑ appetite
Heart rate	Bradycardia	Tachycardia
Skin	Alopecia, thickened skin, dry coat	Usually not the classic finding
Cholesterol	↑	↓
GI signs	Possible constipation	Vomiting/diarrhea
Water intake	May vary	↑ Polydipsia + polyuria
Main screening hormone	TT4 + TSH	TT4
Additional test	FT4 by equilibrium dialysis	FT4, T3 suppression, scintigraphy

GOITROGEN SUMMARY

Goiter:

↓ **Thyroid hormone production** → thyroid enlargement

Goitrogens:

Goitrin → interferes with iodine organification

Thiocyanate → interferes with iodide trapping

Examples of goitrogenic plants:

Cabbage, kale, rutabaga, turnip, rapeseed

Excess iodine:

Can sometimes overcome **thiocyanate**, but is less effective against **goitrin**.

CANINE HYPOTHYROIDISM: MOST IMPORTANT POINTS

Most common primary cause:

Lymphocytic thyroiditis

Classic signs:

Lethargy + obesity + symmetrical alopecia + dry/dull hair + poor hair regrowth

Important laboratory findings:

Hypercholesterolemia (~75%)

Anemia (~25–30%)

Mild hyponatremia (~30%)

Possible ↑ CK

Important diagnostic rule:

Low TT4 alone does NOT prove hypothyroidism.

Consider:

- Breed
- Age
- Illness
- Drugs
- Euthyroid sick syndrome

Practical testing:

TT4 + TSH → FT4 by equilibrium dialysis

FELINE HYPERTHYROIDISM: MOST IMPORTANT POINTS

Most common feline endocrinopathy

Most common pathology:

Adenomatous hyperplasia → autonomous thyroid tissue

Classic clinical picture:

Polyphagia + weight loss + hyperactivity + tachycardia

Also:

Vomiting + diarrhea + PU/PD

Important laboratory findings:

↑ **BUN** may occur

Creatinine may remain normal because GFR is increased

↓ **urine specific gravity**

↑ **ALT/AST** in ~80–90%

↓ **cholesterol**

Diagnosis:

TT4 → FT4 for early/occult cases

T3 suppression test → borderline cases

Scintigraphy → identifies hyperfunctional thyroid tissue